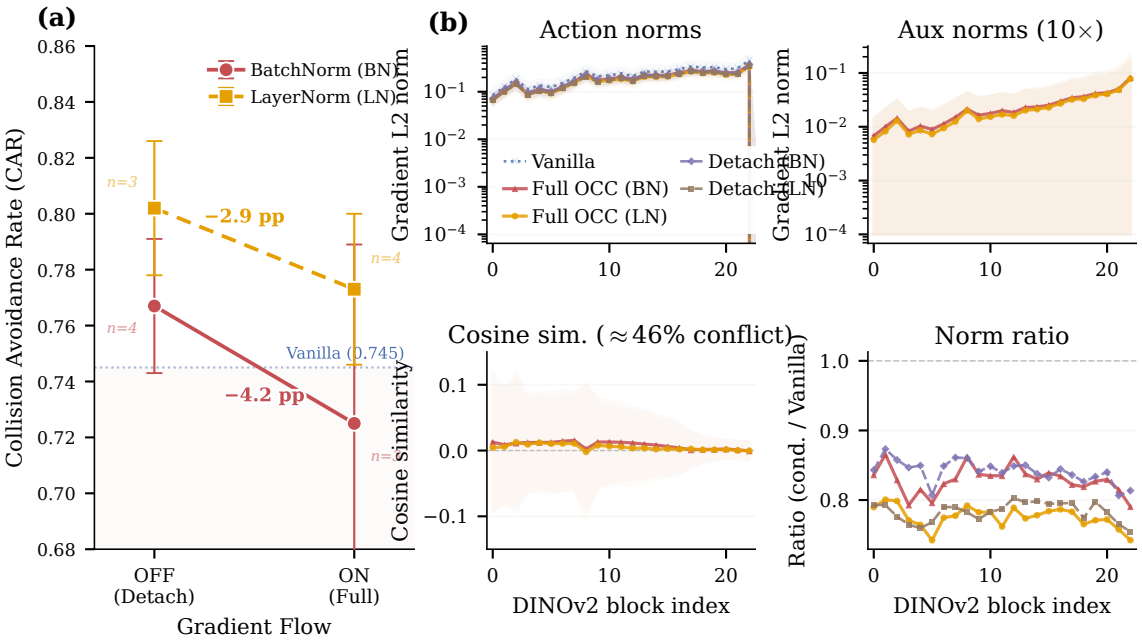
(a)
BatchNorm (BN)
LayerNorm (LN)
Collision Avoidance Rate (CAR)
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0.68
n=3
n=4
−2.9 pp
n=4
Vanilla (0.745)
−4.2 pp
n=3
OFF
(Detach)
ON
(Full)
Gradient Flow
(b)
Action norms
Gradient L2 norm
Vanilla
Full OCC (BN)
Full OCC (LN)
Detach (BN)
Detach (LN)
Aux norms (10×)
Gradient L2 norm
Cosine sim. (≈46% conflict)
Cosine similarity
DINOv2 block index
Norm ratio
Ratio (cond. / Vanilla)
DINOv2 block index